Problem 5

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Problem 1 *Suppose a function f satisfies the following conditions:*

(a) $f(0) = 0$ *and* $f'(-4) = f'(2) = f'(10) = 0$

(b) $\lim_{x \to 6} f(x) = -\infty$, *and* $\lim_{x \to +\infty} f(x) = 6$

(c) $f'(x) < 0$ *on* $(-\infty, -4)$, $(2, 6)$, *and* $(10, +\infty)$

(d) $f'(x) > 0$ *on* $(-4, 2)$, *and* $(6, 10)$

(e) $f''(x) > 0$ *on* $(-\infty, 0)$, *and* $(14, +\infty)$

(f) $f''(x) < 0$ *on* $(0, 6)$, *and* $(6, 14)$

(a) *List the **interval(s)** where the function f is **both increasing** and **concave UP**.*

(b) *List the **interval(s)** where the function f is **both increasing** and **concave DOWN**.*

(c) *List the **interval(s)** where the function f is **both decreasing** and **concave UP**.*

(d) *List the **interval(s)** where the function f is **both decreasing** and **concave DOWN**.*

(e) *List the **x-coordinates** at which f has a **local minimum**. Write "none" if appropriate.*

(f) *List the **x-coordinates** at which f has a **local maximum**. Write "none" if appropriate.*

(g) *List the* ***x-coordinates*** *of all* ***inflection points*** *of f. Write "none" if appropriate.*

(h) *Sketch the graph of f.*

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